

DeepCausality Project 2026 Update

The year 2026 continued to expand the foundation that was renewed late in the prior year. In 2025, the causal monad was introduced and formalized, and the effect propagation process was subsequently defined and implemented as the foundation for dynamic causality. The full formalization of the effect propagation process remains ongoing research because of the non-trivial mathematics involved, which spans, among others, applied category theory, abstract algebra, number theory, and set theory. Despite these difficulties, approximately half of the foundation has been formalized and Lean-verified. The other half primarily concerns the subsumption of other formalisms and the expansion into the formal foundations of quantum causal models.

In 2026, the Center for Dynamic Causality was formed to support this ongoing research and to drive the practical application of dynamic causality. The center set forth three research areas:

- 1) Foundational Dynamic Causality
- 2) Counterfactual Fluid Dynamics
- 3) DeepBrain: Dynamic knowledge representation as the sibling of dynamic causality

In support of all these areas, the DeepCausality project added:

- Float106 – a custom high-precision float type that serves as a stop-gap until f128 arrives in Rust
- Precision as a parameter throughout the unified math, implemented via a custom algebraic type system used across all DeepCausality crates. As a result, all scientific simulations can run at every supported floating-point precision level (f32, f64, Float106) by changing a single type alias. Once Rust lands native f16 and f128 floating-point types, these will be supported as well.
- Implementation of lattice gauge field theory as preparation for dynamic causality in advanced sciences such as general relativity
- A new project website was built and launched
- Miri verification was added to CI to safeguard against undefined behavior

Foundational dynamic causality was advanced in 2026 in the following ways:

A state-of-the-art BRCD algorithm for causal discovery, first published at ICML 2026, was implemented.

The BRCD algorithm was wired into the Causal Discovery Language on June 8 (PR #616), with a learn-once, rank-many CPDAG cache. Later in the month it gained parallel evaluation across candidates, opt-in near-linear MAP-config pruning, and a DAG sampling module with a polynomial-time Clique-Picking AMO counter.

The Causal Arrow generalization (PR #613) generalized the causal monad to an Arrow, introducing Morphism and Endomorphism in the underlying algebra. The CausalFlow DSL landed on June 5, giving the monad's power without its type-level ceremony.

Automatic differentiation was added as forward-mode autodiff over the dual number type, together with numeric integration operators (Euler, RK4, quadrature). The dual number crate and the subsequent calculus crate were published to crates.io. The fast Fourier transform and its inverse were implemented and published in `deep_causality_fft` on crates.io.

Counterfactual fluid dynamics was advanced in 2026 in the following ways:

The combined work on the causal arrow, automatic differentiation, and the FFT, together with the unified math, has led to `deep_causality_cfd`, a new crate (PR #629, PR #637) dedicated to state-of-the-art fluid dynamics fused with dynamic causality. The prior foundational work in advanced field theory has led to the insight that multi-physics can be coupled through a shared field while preserving the invariance of each physics kernel. A quantized tensor-network stack was added to the existing tensor library.

A QTT solver was added to the CFD project in June. It comprises a quantized-tensor-train bridge (codec, finite-difference operators, linear rollout), a 2-D incompressible Navier-Stokes solver, an immersed body with surface observables, and the Tier-B compressible marcher with 3-D operators and a MetricProvider seam that makes body fitting data rather than a code path.

Counterfactual fluid dynamics was then illustrated with a retropropulsion code example that demonstrates the orbital re-entry of a space rocket in reverse: gliding through a plasma blackout, transitioning through multiple governing regimes (supersonic, transonic,

subsonic, GNSS denied, GNSS available), and simulating counterfactuals to optimize its retropropulsion booster ignition for a precision landing.

Quantum causal models were initially not planned for 2026. However, the PhD thesis and related publications of Robin Lorenz (Oxford, 2020) were discovered by accident, and his formal method of using monads for quantum causal models proved nearly identical to the use of arity-5 monads in DeepCausality for causal composition. The Dynamic Quantum Causality project was therefore initiated ad hoc in July 2026.

The Dynamic Quantum Causality project has already implemented the core of Lorenz causal models embedded in the effect propagation process, has implemented the Haruna gates (Haruna, Osaka, 2025), and is now exploring a novel bridge between quantum causal models and dynamic error codes via the Haruna gates. This became possible because the Haruna formalism is rooted in lattice gauge field theory, which was implemented earlier this year in the geometric algebra crate in anticipation of its use in advanced science. Originally, the idea was to explore dynamic causal structures in condensed matter physics, but quantum causal models were such an obvious fit that the new project was warranted.

DeepBrain: Dynamic knowledge representation

Originally, the DeepBrain project was designed to address the ongoing documentation challenge of the DeepCausality project. Unlike other projects in the field, the DeepCausality project carries a very high degree of deeply integrated cross-domain knowledge. This becomes apparent when looking at the type system used across the uniform math. Unlike conventional mathematical libraries, the DeepCausality project does not lean on the default type system provided by the Rust programming language. Instead, the correct algebraic abstractions are modeled by hand, and mathematical operations are typed against the corresponding algebraic categories, such as domains, semirings, fields, or complex fields. As a result, something as simple as a linear algebra library suddenly touches on applied category theory, abstract algebra, linear algebra, and type theory.

The documentation challenge is illustrated by two specific problems. First, the origin of a concept needs to be attributed, and in many cases that origin is a publication or an internal note. The origin then needs to be linked to the specification of a derived feature, which in turn needs to be linked back to the actual implementation of that specification. Second,

and this is critical, the usage of that implementation across the entire monorepo needs to be tracked as well.

This provenance is critical for refactoring, because any modification of the underlying algebraic type system directly affects one third of the published crates, and implicitly at least three quarters, if not more. The unified math is the fabric that holds the entire project together, and beneath it lies its algebraic foundation. Precise tracing from idea to specification to implementation and downstream usage is therefore paramount to a gradual transition to non-breaking refactoring. The DeepBrain project was specifically designed around this use case, with the foundational idea that this lineage evolves continuously and that the underlying knowledge graph therefore needs to be capable of representing the dynamic evolution of that lineage.

The DeepBrain project was temporarily deprioritized in favor of the emerging Quantum Causal project.

However, because of the ongoing integration of DeepCausality into the Service Radar project, renewed interest in the DeepBrain project came from the Service Radar team. Specifically, there was a very explicit need for dynamic ontologies as part of their unified networking information system. An internal feasibility study was commenced and deemed the project feasible with a slightly reduced scope, but DeepBrain has not yet commenced.

Looking ahead to the rest of the year, the roadmap is clear.

- Advance the Service Radar integration
- Advance the Quantum Causal project
- Continue the formalization
- Implement the first stage of the DeepBrain project

Marvin Friedrich Lars Hansen

Director, Center for Dynamic Causality

Project Lead, DeepCausality Project

Tuesday, August 25, 2026

Appendix A: A list of all crates published in 2026

The DeepCausality project published a total of 11 new crates in 2026:

- `deep_causality_calculus` (Automatic differentiation)
- `deep_causality_fft` (Fast Fourier transform)
- `deep_causality_par` (Utilities for multi-core programming)
- `deep_causality_cfd` (Counterfactual fluid dynamics)
- `deep_causality_file` (IO monad and functional file handling)
- `deep_causality_algebra` (Slimmed-down version of pure algebra)
- `deep_causality_num_complex` (All three complex number sets)
- `deep_causality_num_dual` (Dual number set for automatic differentiation)
- `deep_causality_quantum` (Quantum causal models and Haruna gates)
- `deep_causality_num_rational` (Rational number set)
- `deep_causality_linear` (Linear algebra)